DEFINICIJE 1

Navadna diferencialna enačba 1. reda $\dot{x} = F(x,t)$

Skalarna linearna enačba 1. reda $\dot{x} = f(t)x + g(t)$

Integralska krivulja $\dot{\gamma}(t) = \vec{f}(\gamma(t))$

Eksistenčni izrek $\cdot\, x_r(0) = x_0$ $\cdot\, \dot{x} = F(t,x)$

- F lipschitzova na na $\vec{x}$ z L
- $F \leq M$ na $[t_0-a, t_0+a] \times B_b$

Potem rešitev obstaja in je enolična na

$[t_0-\alpha, t_0+\alpha] \longrightarrow B_b$, kjer $\alpha \leq \max\{\frac{1}{2M}, \frac{1}{L}\}$

Prvi integral u za $y' = F(x,y)$, če za $\forall$ rešitev $y(x)$

velja $u(x, y(x)) \equiv C$

Pfafova enačba $P(x,y)dx + Q(x,y)dy = 0$

Integrajoči množitelj $\lambda(x,y)$, da je $\lambda \cdot F$ potencialno

Lagrange d'Alembertova formula $y = x \cdot \varphi(y') + \psi(y')$

Clairotova enačba $y = xy' + \psi(y')$

Ovojnica družine krivulj $F(x,y;c) \equiv 0$ je $c \mapsto (x(c), y(c))$

za katero velja $\forall c$. $(x'(c), y'(c)) \parallel (\dot{x}(c,t), \dot{y}(c,t))$

DEFINICIJE 2

Picardova preslikava $\mathcal{A}(\vec{x}) = \vec{x}_0 + \int_0^t F(\tau, x(\tau))\, d\tau$

Peanov izrek • $F(t,\vec{x}) : [t_0-a, t_0+a] \times B_{b,x_0} \longrightarrow \mathbb{R}^n$ zvezna

Potem $\exists \alpha$ da obstaja vsaj ena rešitev $\dot{\vec{y}} = F(t,\vec{y}), \vec{y}(t_0) = \vec{x}_0$

Tok vektorskega polja $\Phi_t(\vec{x}_0) = \vec{y}(t, x_0)$ za $\vec{y}$ rešitev problema $\dot{\vec{y}} = F(t,\vec{y})$ $\quad y(t_0) = x_0$

Infentizimalni generator $\{\Phi_t\}$ je polje, da velja $\frac{d}{dt}\Big|_{t=0} \Phi_t(x) = F(x)$

Kompletno polje F, če za $\forall x_0 \in \Omega$ je rešitev problema $\dot{\vec{x}} = F(\vec{x})$, $\vec{x}(0) = \vec{x}_0$ definirana za $\forall t \in \mathbb{R}$

Hamiltonski sistem $\Omega \subseteq \mathbb{R}^{2n}$ $\quad \Omega = \{(q_1, \dots q_n, p_1, \dots p_n)\}$

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Liuovillova enačba • Φ_t tok vek. polja F na $\Omega \subseteq \mathbb{R}^n$

$$\frac{d}{dt}\det(J\Phi_t(\vec{a})) = \operatorname{div}(F(\Phi_t(\vec{a}))) \cdot \det(J\Phi_t(\vec{a}))$$

Oziroma $\det(J\Phi(\vec{a})) = e^{\int \operatorname{div}(F(\tau,\vec{a}))d\tau}$

Hamiltonsko polje $X_H = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \nabla H$

Nepopolni eliptični integral 1. vrste $F(z,k) = \int_0^a \frac{d\tau}{\sqrt{1-k\sin^2(\tau)}}$

Jacobijeva amplituda $\operatorname{am}(z,k) = F^{-1}(z,k)$

Poissonov oklepaj $\{f,g\} = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial q_i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q_i} \right) = X_f \cdot \nabla g$

Liejev produkt vektorskih polij

- antisimetričnost $[F,G] = -[G,F]$
- Jacobijeva identiteta

$$[E,[F,G]] + [G,[E,F]] + [F,[G,E]] = 0$$

Simetrija Hamiltonskega sistema (H,Ω) je difeomorfizem $S: \Omega \longrightarrow \Omega$, da γ rešitev za (H,Ω) $\quad \gamma(t) = (p(t), q(t))$ $\gamma(0) = x_0$, potem je $S(\gamma(t))$ rešitev za (H,Ω) z začetnim pogojem $S(\gamma(0)) = S(x_0)$

DEFINICIJE 3

$$\dot{\vec{x}} = A(t)\vec{x}$$

Fundamentalna matrika $\Phi_{t_0,t}$ sistema $\dot{\vec{x}} = A(t)\vec{x}$ je

$\Phi = [\vec{x}_1 \ \dots \ \vec{x}_n]$ kjer so $\vec{x}_i$ baza rešitev. $\vec{x}_i(t_0) = \vec{e}_i$

Eksponencijacija matrike $e^A = I + \sum_{n=1}^{\infty} \frac{1}{n!} A^n$

BCH formula $e^A e^B = e^C \rightsquigarrow C = \ln(e^A e^B)$

Dysonova vrsta $\Phi_t = I + \sum_{n=1}^{\infty} \int_{O_n(t)} A(\tau_1)A(\tau_2)\dots A(\tau_n)\, d\vec{\tau}$

Liouvillova formula $\dot{\det}(\Phi_t) = \mathrm{sl}(A(t))\det(\Phi_t)$

$$\det(\Phi_t) = \det(\Phi_{t_0})\, e^{\int_{t_0}^{t} \mathrm{sl}(A(\tau))\,d\tau}$$

Determinanta wronskega $W(t) = \det(\Phi_t)$

DEFINICIJE 4

Lagrangeev funkcional $\mathcal{L}(y) = \int_a^b L(x, y(x), y'(x))\,dx$

Šibki odvod (smerni, Gateauxov) $P: U \to V$

$U^G_{u_0}(v) = \lim_{t\to 0} \frac{1}{t}\left(P(u_0+tv) - P(u_0)\right) = \frac{d}{dt}\Big|_{t=0} P(u+tv)$

Krepki odvod (Frechetov) $P: U \to V$ $A: U \to V$ linearna, omejena. $P(u_0+v) = P(u_0) + A(v) + \sigma(u_0, v)$; $\lim_{\|v\|\to 0} \frac{\|\sigma\|}{\|v\|} = 0$

Dopustna funkcija $v \in \{u \in U.\ u(a) = A,\ u(b) = B\}$

Dopustna variacija $v \in \{u \in U.\ u(a) = 0,\ u(b) = 0\}$

Euler-Lagrangeeva enačba $\frac{\partial L}{\partial y}(x, y(x), y'(x)) - \frac{d}{dx}\frac{\partial L}{\partial y'}(\dots) = 0$

Vezani ekstrem: Pogoj $\Phi(u) = \ell \in \mathbb{R}$, ekstrem $\mathcal{L}$

KAJ MORAM ZNATI

- Kako vemo, da je rešitev sistema globalno definirana
- Dysonova vrsta
- Liouvillova formula
- Vezani ekstremi
- euler-lagrangova enačba
- tangentni prostor je enak jedru vezi
- eksistenčni izrek
- Variacijski račun
- odvod $\mathcal{L}$
- dopustne variacije
- tok in lastnosti
- primeri za eksistenčni izrek
- Fundamentalna matrika

Eksistenčni izrek

- $F(t,\vec{x}) : [t_0-a, t_0+a] \times B_b \longrightarrow \mathbb{R}$ zvezna, Lipschitzova na drugi spremenljivki s konstanto L.
- $\dot{\vec{y}} = F(t,\vec{y}) \quad \vec{y}(t_0) = \vec{x}_0 \in B_{\frac{b}{2}}$

Potem obstaja natanko ena rešitev problema

$$\vec{y} : [t_0-\alpha, t_0+\alpha] \longrightarrow B_b \qquad \alpha \leq \max\{\tfrac{b}{2M}, \tfrac{1}{L}\}$$

$$M = \max\{F(t,\vec{x}) ; (t,\vec{x}) \in [t_0-a, t_0+a] \times B_b\}$$

Dokaz:

$$\vec{x}(t) = \vec{x}_0 + \int_{t_0}^{t} F(\tau, \vec{x}(\tau))d\tau$$

$$\mathcal{A}(\vec{x}) = \vec{x}(t_0) + \int_{t_0}^{t} F(\tau, \vec{x}(\tau))d\tau \qquad \mathcal{A} : V \longrightarrow V$$

$\vec{x}$ je rešitev $\Leftrightarrow \vec{x}(t) = \mathcal{A}(\vec{x}(t)) \Leftrightarrow \vec{x}$ je fiksna točka.

Taka točka obstaja, ko je $\mathcal{A}$ skrčitev

$$P_{a,b} = \{(t,\vec{x}) \in \mathbb{R} \times \mathbb{R}^n ; |t-t_0| < a, \|\vec{x}-\vec{x}_0\| \leq b$$

$$E = \{\vec{y} : \Omega \rightarrow \mathbb{R} ; \vec{y}(t_0,\vec{x}) = \vec{x} ; \|\vec{y}(t,\vec{x}) - \vec{x}_0\| \leq b \ \forall t,x \in P_{a,\frac{b}{2}}\}$$

$$\Omega = [t_0-\alpha, t_0+\alpha] \times B_{\frac{b}{2}}$$

$\mathcal{A}(E) \subseteq E$

1) $\mathcal{A}(\vec{y}(t_0,\vec{x})) = \vec{x} \qquad \mathcal{A}(\vec{y}) - \vec{x} = \int_{t_0}^{t_0} F(\tau, y(\tau,\vec{x}))d\tau = 0$

2) $\|\mathcal{A}(\vec{y}(t,\vec{x})) - \vec{x}_0\|_\infty \leq \|\vec{x} - \vec{x}_0\| + \int_{t_0}^{t} F(\tau, y(\tau,\vec{x}))d\tau$

$$\leq \frac{b}{2} + \alpha M \leq b \quad \text{za} \quad \alpha \leq \frac{b}{2M}$$

Ali je $\mathcal{A}$ na E skrčitev?

$$\|\mathcal{A}(\vec{y}_1) - \mathcal{A}(\vec{y}_2)\|(t,\vec{x}) = \|\int_{t_0}^{t} F(\tau,y_1) - F(\tau,y_2)d\tau\|_\infty \leq$$

$$\leq \|\int_{t_0}^{t} L\|y_1(\tau,x) - y_2(\tau,x)\|d\tau\| \leq \alpha L \|y_1 - y_2\|_\infty$$

$$\alpha L < 1 \rightsquigarrow \alpha < \frac{1}{L}$$

Torej rešitev obstaja

Zvezna odvisnost od začetne točke

$$y_1(t,x_1) \quad y_2(t,x_2) \quad ; \|x_1 - x_2\| < \delta \qquad = y_1(t,x_1) - y_2(t,x_2)$$

$$\|\mathcal{A}(t, y_1(t,x_1)) - \mathcal{A}(t, y_2(t,x_2))\|_\infty \leq$$

$$\leq \|x_1 - x_2\| + \|\int_{t_0}^{t} L\|y_1(t,x_1) - y_2(t,x_2)\|dt\| \leq$$

$$\leq \delta + \|\int_{t_0}^{t} L\|y_1 - y_2\|_\infty dt = \delta + L\|y_1 - y_2\|_\infty \cdot \alpha$$

$$\|y_1 - y_2\|_\infty (1 - L\alpha) \leq \delta$$

$$\|y_1 - y_2\| \leq \frac{\delta}{L\alpha}$$

Tok vektorskega polja

$$\dot{\Phi}_t(x) = F(t, \Phi_t) \quad / \frac{d}{dx}$$

$$(D_x \dot{\Phi}_{t,t_0})(h) = (D_x F)(t, \Phi_{t,t_0}) \cdot (D_x \Phi_{t,t_0})(h)$$

$$\rightsquigarrow \frac{d}{dt}(D_x \Phi) = D_x F(t, \Phi_t) \cdot D_x \Phi_{t,t_0} \qquad D_x \Phi_{t_0} = I$$

Pomembni lastnosti:

1) $\Phi_{t_0,t_0}(\vec{x}) = \vec{x}$

2) $\Phi_{t,s} \circ \Phi_{s,t_0} = \Phi_{t,t_0}$

Avtonomen sistem $\rightsquigarrow \Phi_{t,t_0} = \Phi_{t+\tau, t_0+\tau} \Rightarrow \Phi_{t,t_0} = \Phi_{s,0}$

$\rightsquigarrow$ 1) $\Phi_0 = I$

2) $\Phi_{s+t} = \Phi_s \circ \Phi_t$

$\Rightarrow \Phi_t^{-1} = \Phi_{-t} \Rightarrow \Phi_t$ je difeomorfizem

Izrek: Liouvillova enačba

Φ_t tok avtonomnega polja $F(\vec{x})$ $\quad J\Phi = \det(D\Phi)$

Potem $\frac{d}{dt} J(\Phi_t(\vec{x})) = \operatorname{div}(F(\Phi_t(\vec{x})))\, J(\Phi_t(\vec{x}))$

Dokaz: (za $n=2$)

Označimo $\quad \Phi_t \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a(t,x,y) \\ b(t,x,y) \end{bmatrix}$

$$\dot{\Phi}_t \begin{bmatrix} x \\ y \end{bmatrix} = F\left(\Phi_t \begin{bmatrix} x \\ y \end{bmatrix}\right) \rightsquigarrow \begin{bmatrix} \dot{a} \\ \dot{b} \end{bmatrix} = F\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) \Big/ D_x$$

$$D_{\vec{x}}\left(\partial_t \begin{bmatrix} a \\ b \end{bmatrix}\right) = D_{\vec{x}}\left(\begin{bmatrix} f(a,b) \\ g(a,b) \end{bmatrix}\right)$$

$$\left(D_{\vec{x}} \begin{bmatrix} a \\ b \end{bmatrix}\right) = \begin{bmatrix} \partial_x a & \partial_y a \\ \partial_x b & \partial_y b \end{bmatrix}$$

$$J = \partial_t \det\left(D_x\left(\Phi_t \begin{bmatrix} x \\ y \end{bmatrix}\right)\right) = \partial_t \det \begin{bmatrix} \partial_x a & \partial_y a \\ \partial_x b & \partial_y b \end{bmatrix} =$$

$$= \partial_t (\partial_x a \cdot \partial_y b - \partial_y a \cdot \partial_x b) =$$

$$= \partial_t \partial_x a \cdot \partial_y b + \partial_x a\, \partial_t \partial_y b - (\partial_t \partial_y a \cdot \partial_x b + \partial_y a \cdot \partial_t \partial_x b) =$$

$$= \partial_x \partial_t a \cdot \partial_y b - \partial_y \partial_t a \cdot \partial_x b + \partial_x a \cdot \partial_y \partial_t b - \partial_y a \cdot \partial_x \partial_t b =$$

$$\begin{vmatrix} \partial_x \dot{a} & \partial_y \dot{a} \\ \partial_x b & \partial_y b \end{vmatrix} + \begin{vmatrix} \partial_x a & \partial_y a \\ \partial_x \dot{b} & \partial_y \dot{b} \end{vmatrix}$$

Ker velja $\begin{bmatrix} \dot{a} \\ \dot{b} \end{bmatrix} = F\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = \begin{bmatrix} f(a,b) \\ g(a,b) \end{bmatrix}$

je $\quad \dot{a} = f(a,b) \rightsquigarrow \partial_x \dot{a} = f_a \partial_x a + f_b \partial_x b$

$$J = \begin{vmatrix} f_a \partial_x a + f_b \partial_x b & f_a \partial_y a + f_b \partial_y b \\ \partial_x b & \partial_y b \end{vmatrix} +$$

$$+ \begin{vmatrix} \partial_x a & \partial_y a \\ g_a \partial_x a + g_b \partial_x b & g_a \partial_y a + g_b \partial_y b \end{vmatrix} =$$

$$= f_a (\partial_x a \cdot \partial_y b - \partial_y a \partial_x b) + f_b (\partial_x b \partial_y b - \partial_y b \partial_x b)$$

$$+ g_a (\partial_x a \partial_y a - \partial_y a \partial_x a) + g_b (\partial_x a \partial_y b - \partial_y a \partial_x b) =$$

$$= \begin{bmatrix} f_a \\ g_b \end{bmatrix} \cdot \begin{vmatrix} \partial_x a & \partial_y a \\ \partial_x b & \partial_y b \end{vmatrix} = \operatorname{div} F\left(\Phi_t \begin{bmatrix} x \\ y \end{bmatrix}\right) \cdot \det\left(D_x \Phi_t \begin{bmatrix} x \\ y \end{bmatrix}\right)$$

■

Trditev fundamentalna matrika Φ je obrnljiva. ($\dot{\Phi} = A(t)\Phi$

Dokaz: Recimo da $\Phi_{t_0,t_1}\vec{a} = \vec{0}$

$$\vec{y}(t) = \Phi_{t_0,t}\vec{a} = \sum_i \vec{a}_i \vec{X}_i(t)$$

$$\vec{y}(t_1) = \vec{0}$$

$\vec{y}$ z začetnim pogojem $\vec{y}(t_1) = 0$

reši enačbo $\dot{y} = F(t,y)$, $y(t_1) = 0$

to reši tudi $z \equiv 0$, torej po eksistenčnem izreku je $y \equiv 0$

torej $\vec{a} \equiv 0$ torej ni netrivialnega jedra.

■

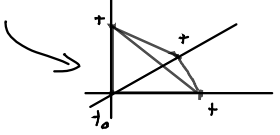
Izrek: Dysonova vrsta

- $A : [t_0, t] \longrightarrow \mathrm{Mat}_{n\times n}$ omejena po operatorski normi (norma, kjer $\|I\|=1$)

Potem je fundamentalna matrika enačbe $\dot{\vec{x}} = A(t)\vec{x}$

$$\Phi_{t,t_0} = I + \sum_{n=1}^{\infty} \int_{\Delta_n(t)} A(\tau_1)A(\tau_2)\dots A(\tau_n)\,d\tau$$

kjer je $\Delta_n(t) = \{(\tau_1,\dots \tau_n) \subseteq \mathbb{R}^n;\ t_0 \leq \tau_n \leq \dots \leq \tau_n \leq t\}$



Dokaz: Poglejmo kdaj velj. $\dot{\Phi} = A(t)\Phi$ oziroma

$$\Phi = I + \int_{t_0}^{t} A(\tau)\Phi(\tau)\,d\tau$$

Rekurzivno kliče enačba sama sebe:

$$\Phi = I + \int_{t_0}^{t} A(\tau_1)\Big(I + \int_{t_0}^{\tau_1} A(\tau_2)\Big(I + \int_{t_0}^{\tau_2} A(\tau_3)(I + \dots)\dots d\tau_3\, d\tau_2\, d\tau_1$$

$$= I + \int_{t_0}^{t} A(\tau_1)\,d\tau_1 + \int_{t_0}^{t} d\tau_1 \int_{0}^{\tau_1} A(\tau_2)\,d\tau_2 + \int_{t_0}^{t} d\tau_1 \int_{t_0}^{\tau_1} d\tau_2 \int_{t_0}^{\tau_2} A(\tau_3)\,d\tau_3 \dots$$

$$= I + \sum_{n=1}^{\infty} \int_{\Delta_n(t)} A(\tau_1)\dots A(\tau_n)\,d\tau_1 \dots d\tau_n = \text{vrsta}$$

To je izpeljava formule.
Preverimo ali je res rešitev

$$\dot{\Phi}_t = \partial_t \sum_{n=1}^{\infty} \int_{\Delta_n(t)} \dots = \partial_t \sum_{n=1}^{\infty} \int_{t_0}^{t} d\tau_1 \int_{t_0}^{\tau_1} d\tau_2 \int \dots A(\tau_1)\dots A(\tau_n)\,d\tau_n$$

$$\partial_t \int_{t_0}^{t} f(\tau)\,d\tau = f(t)$$

$$= \sum_{n=1}^{\infty} \int_{t_0}^{t} d\tau_2 \int_{t_0}^{\tau_2} \dots A(t)A(\tau_2)\dots A(\tau_n)\,d\tau_n =$$

$$= A(t)\Big(I + \sum_{n=2}^{\infty} \int_{t_0}^{t} d\tau_2 \int_{t_0}^{\tau_2} \dots A(\tau_2)\dots A(\tau_n)\,d\tau_n\Big) =$$

$$= A(t)\Big(I + \sum_{n=1}^{\infty} \int_{t_0}^{t} d\tau_1 \int_{t_0}^{\tau_2} \dots A(\tau_1)\dots A(\tau_n)\,d\tau_n$$

preimenujemo spremenljivke in v vrsti prestavimo $n \mapsto n-1$

$$= A(t)\cdot\Phi_t$$

Preveriti moramo, ali lahko zamenjamo ∂_t in Σ in ali vrsta Φ_t sploh obstaja.

$$\|\Phi_t\| \leq 1 + \sum_{n=1}^{\infty} \int_{t_0}^{t} d\tau_1 \int_{t_0}^{\tau_1} \dots \|A(\tau_1)\|\,\|A(\tau_2)\|\dots\|A(\tau_n)\|\,d\tau_n \leq$$

$$\leq 1 + \sum_{n=1}^{\infty} \int_{\Delta_n(t)} M^n = 1 + \sum_{n=1}^{\infty} \frac{(t-t_0)^n}{n!} M^n =$$

$$= \sum_{n=0}^{\infty} \frac{1}{n!}((t_0-t)M)^n = e^{(t_0-t)M} < \infty$$

Torej Φ_t obstaja.

Ali lahko zamenjamo ∂_t in Σ v dokazu?

$$\sum_{n=1}^{\infty} \int_{\Delta_n(t)} A(\tau_1)\dots A(\tau_n)\,d\tau \leq \sum_{n=1}^{\infty} \frac{1}{n!}((t-t_0)M)^n < \infty$$

Torej po Weierstrassovem M-testu je to enakomerna konvergenca, torej $\partial_t \Sigma \dots = \Sigma\, \partial_t \dots$

Izrek: Formula, ko A_j komutirajo

Če velja $\forall t_1, t_2 \in [0,b]$

$$[A(t_1), A(t_2)] = A(t_1)A(t_2) - A(t_2)A(t_1) = 0$$

Potem za $\dot{\Phi}_t = A(t)\Phi_t$, $\Phi_0 = I$ velja rešitev

$$\Phi_t = e^{\int_0^t A(\tau)d\tau}$$

Dokaz:

$$\Phi_t = I + \sum_{n=1}^{\infty} \int_{\Delta_n(t)} A(\tau_n)\dots A(\tau_n)d\tau$$

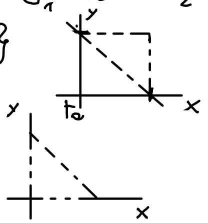
Definirajmo

$$\Delta_n^\sigma(t) = \{(\tau_1, \dots \tau_n) \in \mathbb{R}^n ; t_0 < \tau_{\sigma(1)} < \dots < \tau_{\sigma(n)} < t\}$$

za $n=2$ imamo dve permutaciji: $\sigma_1 = id$ $\sigma_2 = (1\ 2)$

$$\Delta_n^{\sigma_1}(t) = \{(x,y) ; t_0 < x < y < t\}$$

$$\Delta_n^{\sigma_2}(t) = \{(x,y) ; t_0 < y < x < t$$



$$\Delta_2^{\sigma_1}(t) \sqcup \Delta_2^{\sigma_2}(t) = \square_2(t)$$

$$\square_n(t) = \{(\tau_1, \dots \tau_n) \in \mathbb{R}^n ; \forall i,j .\ i \neq j \Rightarrow \tau_i \neq \tau_j\}$$

τ brez diagonal

Podobno $\coprod_{\sigma \in S_n} \Delta_n^\sigma(t) = \square_n(t)$

Ker je $\prod A(\tau_i) = \prod A(\tau_{\sigma(i)})$ (ker matrike komutirajo)

velja $\int_{\Delta_n^\sigma} A(\tau_{\sigma(1)}) \dots A(\tau_{\sigma(n)}) d\tau = \int_{\Delta_n} A(\tau_1) \dots A(\tau_n) d\tau$

$$\Rightarrow \int_{\square_n(t)} A(\tau_1) \dots A(\tau_n) d\tau = \sum_{\sigma \in S_n} \int_{\Delta_n^\sigma(t)} A(\tau_1) \dots A(\tau_n) d\tau =$$

$$= n! \int_{\Delta_n(t)} A(\tau_1) \dots A(\tau_n) d\tau$$

$$\Rightarrow \Phi_t = I + \sum_{n=1}^{\infty} \frac{1}{n} \int_0^t A(\tau_1)d\tau_1 \int_0^t A(\tau_2)d\tau_2 \dots \int_0^t A(\tau_n)d\tau_n =$$

$$= I + \sum_{n=1}^{\infty} \frac{1}{n!} \left(\int_0^t A(\tau)d\tau \right)^n = e^{\int_0^t A(\tau)d\tau} \blacksquare$$

Izrek: Globalnost rešitev

Naj bo $A(t) \leq M$ za $\forall t \in [t_0, t_1]$

Potem je rešitev $\vec{x}(t)$ za $\dot{\vec{x}} = A(t)\vec{x}$ $\vec{x}(0) = \vec{a}$ definirana na celem intervalu $[t_0, t_1]$

Dokaz:

$\vec{x}(t) = \Phi_{t_0,t}\vec{a}$ Ali: Φ_t obstaja povsod?

$$\|\Phi_{t_0,t}\| = \left\|I + \sum_{n=1}^{\infty} \int_{\Delta_n(t)} A(\tau_1)\dots A(\tau_n)\,d\tau\right\| \leq$$

$$\leq 1 + \sum_{n=1}^{\infty} (t-t_0)^n \cdot \frac{1}{n!} M^n = e^{(t-t_0)M} \leq e^{(t_1-t_0)M} < \infty$$

$\blacksquare$

Izrek: Liouvillova formula

Naj bo $\dot{\Phi}_t = A(t)\cdot\Phi_t$

Potem velja $\frac{d}{dt}\det(\Phi_t) = sl(A(t))\cdot\det\Phi_t$

Dokaz:

$$\Phi_t = \begin{bmatrix} x_{11}(t) & \dots & x_{1n}(t) \\ \vdots & & \\ x_{n1}(t) & \dots & x_{nn}(t) \end{bmatrix}$$

$$\det(\Phi_t) = \sum_{\sigma\in S_n} (-1)^{sgn(\sigma)} x_{1\sigma(n)}\dots x_{n\sigma(n)}$$

$$\frac{d}{dt}\det(\Phi_t) = \sum_{\sigma\in S_n}(-1)^{sgn(\sigma)}\sum_{i=1}^{n} x_{1\sigma(n)}\dots\dot{x}_{i\sigma(i)}\dots x_{n\sigma(n)} =$$

$$\dot{\Phi} = A(t)\Phi \rightsquigarrow \dot{x}_{i\sigma(i)} = \sum_{j=1}^{n} a_{ij}x_{j\sigma(i)}$$

$$= \sum_{\sigma\in S_n}(-1)^{sgn\sigma}\cdot\sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}\, x_{1\sigma(1)}\dots x_{j\sigma(i)}\dots x_{n\sigma(n)} =$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}\sum_{\sigma\in S_n}(-1)^{sgn\sigma} x_{1\sigma(1)}\dots x_{j\sigma(i)}\dots x_{n\sigma(n)}$$

imamo σ_1 in $\sigma_2 = (i\,j)\sigma_1$ za $i\neq j$

Potem je $(-1)^{sgn\sigma_1} x_{1\sigma_1(1)}\dots x_{j\sigma_1(i)}\dots x_{i\sigma_1(i)}\dots x_{n\sigma_1(n)} =$

$= -(-1)^{sgn\sigma_2} x_{1\sigma_2(1)}\dots x_{j\sigma_2(j)}\dots x_{i\sigma_2(j)}\dots x_{n\sigma_2(n)}$

Torej vsi členi kjer $i\neq j$ se pokrajšajo

$$= \sum_{i=1}^{n} a_{ii}\sum_{\sigma\in S_n}(-1)^{sgn\sigma} x_{1\sigma(1)}\dots x_{n\sigma(n)} = \sum_{i=1}^{n} a_{ii}\det\Phi_t =$$

$$= sl(A(t))\det\Phi_t$$

■

Trditev: A, B diagonalizabilni in $AB = BA$
Potem obstaja prehodna matrika P, da sta PAP^{-1} in PBP^{-1} diagonalni.

Dokaz:

$Av = \lambda v$

$ABv = BAv = B(\lambda v) = \lambda Bv$

$\Rightarrow Bv$ je lastni vektor za A

$E_\lambda = \{ v \in \mathbb{R}_n ; Av = \lambda v \}$ ... prostor lastnih vektorjev za lastno vrednost λ

$v \in E_\lambda \Rightarrow Bv \in E_\lambda$

Torej B lahko gledamo kot $B_\lambda : E_\lambda \longrightarrow E_\lambda$

B_λ lahko diagonaliziramo, ker lahko diagonaliziramo B

$P_\lambda B_\lambda P_\lambda^{-1} = D_{B_\lambda}$ v bazi $\{ \vec{v}_\lambda^1, \dots, \vec{v}_\lambda^n \}$

$A v_\lambda^i = \lambda v_\lambda \Rightarrow P_\lambda A_\lambda P_\lambda^{-1} = \begin{bmatrix} \lambda & & \\ & \ddots & \\ & & \lambda \end{bmatrix}$

Tako naredimo za $\forall \lambda$ in dobimo

$P = \begin{bmatrix} P_{\lambda_1} & & \\ & \ddots & \\ & & P_{\lambda_k} \end{bmatrix}$ da je $PAP^{-1} = D_A$ in $PBP^{-1} = D_B$

$\blacksquare$

Trditev: če $\mathcal{L}$ v u_0 zavzame lokalni ekstrem, potem je $D_{u_0}\mathcal{L} = 0$

Dokaz:

Recimo da $\forall v \in U.\ \mathcal{L}(u_0+tv) > \mathcal{L}(u_0)$

$$\mathcal{L}(u_0+tv) = \mathcal{L}(u) + (D_{u_0}\mathcal{L})(tv) + o(u_0,v) > \mathcal{L}(u_0)$$

$$(D_{u_0}\mathcal{L})(tv) + o(u_0,tv) > 0 \quad /:t$$

$$(D_{u_0}\mathcal{L})v + \|v\|\frac{o(u_0,tv)}{t\|v\|} > 0 \quad / \xrightarrow[t\to 0]{}$$

$$(D_{u_0}\mathcal{L})(v) + \|v\| \cdot 0 \geq 0 \leadsto$$

$$(D_{u_0}\mathcal{L})(v) \geq 0$$

To velja za $\forall v$, tudi za $-v$

$$\Rightarrow -(D_{u_0}\mathcal{L})v \geq 0$$

$$(D_{u_0}\mathcal{L})v \geq 0 \Rightarrow D_{u_0}\mathcal{L} \equiv 0$$

$\blacksquare$

Ena izpeljava

- $\mathcal{L}$ je krepko odvedljiva
- znamo izračunati šibki odvod

$$(D_u\mathcal{L})(v) = \frac{d}{ds}\Big|_{t=0} \int_a^b L(x, u(x)+sv(x), (u'(x)+sv'(x)))\,dx$$

$$= \int_a^b \left(\frac{dL}{du}\cdot v(x) + \frac{dL}{du'}\cdot v'(x)\right)dx$$

$$\int_a^b \frac{dL}{du'}\,\frac{dv}{dx}\,dx = \underbrace{v\cdot\frac{dL}{du'}\Big|_a^b}_{v(a)=v(b)=0} - \int_a^b \frac{d}{dx}\frac{dL}{du'}\,dx$$

$$\Rightarrow (D_u\mathcal{L})v = \int_a^b \left(\frac{dL}{du} - \frac{d}{dx}\frac{dL}{du'}\right)dx$$

Osnovni izrek variacijskega računa

$u \in \mathcal{C}[a,b]$ poljubna

Če za $\forall \varphi$ testno funkcijo velja $\int_a^b u(x)\varphi(x)\,dx = 0$,

potem je $u \equiv 0$

Dokaz:

Denimo da $\exists c \in [a,b].\ u(c) \neq 0$.

Potem obstaja okolica $[\alpha,\beta]$ za c, da $\forall x \in [\alpha,\beta].\ u(x) \neq 0$

Vzemimo testno funkcijo φ, da je $\mathrm{supp}(\varphi) = (\alpha,\beta)$

Potem je $\int_a^b u(x)\varphi(x)\,dx = \int_\alpha^\beta u(x)\varphi(x)\,dx \neq 0$

ne zamenja predznaka

$\not\equiv 0$

$\times$

$\blacksquare$

Izrek: Euler-Lagrangeeva enačba

- U banachov prostor $\mathcal{C}^1[a,b]$ (ima normo)
- $\mathcal{L}: U \longrightarrow \mathbb{R}$ $\quad u \mapsto \int_a^b L(x,u(x),u'(x))\,dx \quad L \in \mathcal{C}^1$
- $\mathcal{A} = \{u \in U;\ u(a) = A,\ u(b) = B\}$
- $\mathcal{L}|_{\mathcal{A}}$ ima v u_0 lokalni ekstrem $\Leftrightarrow$ u_0 reši

$$\frac{d}{du}L(x,u(x),u'(x)) - \frac{d}{dx}\frac{d}{du'}L(x,u(x),u'(x)) = 0$$

Dokaz:

($\Leftarrow$) sledi iz dejstva $(D_u\mathcal{L})v = \int_a^b \left(\frac{d}{du}L - \frac{d}{dx}\frac{d}{du'}L\right)v(x)$

($\Rightarrow$) Recimo da $(D_u\mathcal{L})v = 0$ za $\forall v$.

Torej velja tudi za v je, kjer $v(a) = v(b) = 0$, to pa so vse testne funkcije.

Torej $\forall \varphi \in$ testna funkcija. $\int_a^b \text{enač}(u_0,x)\cdot\varphi(x)\,dx = 0$

$\Rightarrow$ enačba$(u_0,\cdot) \equiv 0$

Vezani ekstremi

Rešujemo lokalni ekstrem za $\mathcal{L}$ pri pogoju $\Phi(u) \equiv \ell$

$$V = \{u;\ \Phi(u) \equiv \ell\}$$

ekstrem za $\mathcal{L} : V \longrightarrow \mathbb{R}$

u bo lokalni ekstrem, če bo vsaka pot v V imela v u lokalni ekstrem za $\mathcal{L}$

$$\gamma : [-\varepsilon, \varepsilon] \longrightarrow V$$

Recimo, da ima $\mathcal{L}\gamma(\cdot)$ v 0 lokalni ekstrem

$$u_0 = \gamma$$

$$\forall s \in [\varepsilon, \varepsilon].\ \mathcal{L}\gamma(s) \geq \mathcal{L}\gamma(0) = \mathcal{L}u_0$$

$$\frac{d}{ds}\Big|_{s=0} \mathcal{L}\gamma(s) = 0$$

$$\frac{d}{ds}\Big|_{s=0} \mathcal{L}\gamma(s) = \left(D_{\gamma(0)}\mathcal{L}\right)\overbrace{\left(\frac{d}{ds}\Big|_{s=0}\gamma(s)\right)}^{V}$$

$$= \left(D_{u_0}\mathcal{L}\right)(v)$$

Tangentni prostor

$$T_uV = \{ \underbrace{\tfrac{d}{ds}\big|_{s=0} \gamma(s)}_{\gamma'(0)} ;\ \gamma : [\varepsilon, \varepsilon] \to V\ ,\ \gamma(0) = u \}$$

Trditev

- $\Phi : U \to \mathbb{R}$ vez
- $\mathcal{L} : U \to \mathbb{R}$

Velja $\ker D_u \Phi = T_u V$; $V = \{ u \in U ;\ \Phi(u) \equiv \ell \}$

Dokaz: $T_u V \subseteq \ker D_u \Phi$

$$\Phi(\gamma(s)) \equiv \ell \quad / \tfrac{d}{ds}\big|_{s=0}$$

$$(D_{\gamma(0)} \Phi)\, \gamma'(0) = 0$$

$\Rightarrow T_{\gamma(0)} V \subseteq \ker D_{\gamma(0)} \Phi$

$\ker D_u \Phi \subseteq T_u V$

$v \in \ker D_u \Phi$ poljuben

Iščemo $\gamma : [\varepsilon, \varepsilon] \to V$ da je $\gamma'(0) = v$ in $\gamma(0) = u$

Definirajmo $\mathcal{F} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$

$$\xi, \eta \mapsto \Phi(u + \xi v + \eta \cdot w)$$

kjer $v \in \ker D_u \Phi$ in $w \notin \ker D_u \Phi$ $u \in V$

$$\mathcal{F}(0,0) = \Phi(u) = \ell$$

Želeli bomo izraziti $\eta = \eta(\xi)$ s pomočjo izreka o implicitni enačbi v okolici $(0,0)$

$$\frac{d\mathcal{F}}{d\eta}\Big|_{(0,0)} = \frac{d\mathcal{F}(0,\eta)}{d\eta}\Big|_{\eta=0} = \frac{d}{d\eta}\Big|_{\eta=0} \Phi(u + \eta w) =$$

$$= (D_{u+\eta w} \Phi) \cdot w \,\Big|_{\eta=0} = (D_u \Phi) w \neq 0$$

$\rightsquigarrow \eta = \eta(\xi)$ in $\eta(0) = 0$

Definirajmo $\gamma(s) = u + s \cdot v + \eta(s) \cdot w$

Velja $\gamma(0) = u$

Želimo še $\gamma'(0) = a$

$$\gamma'(0) = v + \eta'(0) \cdot w$$

Želimo torej $\eta'(0) = 0$. Preverimo da to drži

$$\mathcal{F}(\xi, \eta(\xi)) = \Phi(u + \xi v + \eta(\xi) \cdot w) = \ell \quad / \tfrac{d}{d\xi}\big|_{\xi=0}$$

$$D_u \Phi (v + \eta'(0) w) = 0$$

$$\underbrace{D_u \Phi v}_{=0} + \eta'(0) \underbrace{(D_u \Phi) w}_{\neq 0} = 0 \Rightarrow \eta'(0) = 0$$

Splošnejša trditev:

- $\mathcal{L} : U \to \mathbb{R}$ funkcional
- $\Phi_i : U \to \mathbb{R}$ za $i \in \{1 \dots n\}$ vezi
- $V = \{u \in U : \forall i. \ \Phi_i(u) = l_i\}$

Potem $T_u V = \bigcap_{i=1}^{n} \ker D_u \Phi_i$

$\subseteq$ isto kot prej

$\supseteq$

Naj bo $\{w_1 \dots w_n\} \subseteq U$ dualna baza za $\{D_u\Phi_1, \dots D_u\Phi_n\}$. Torej $D_u\Phi_i(u_j) = \delta_{ij}$

Definirajmo: $\gamma_\sigma(s) = u + sv + \sigma_1(s)w_1 + \dots + \sigma_n(s)w_n$

kjer $v \in \ker D_u$.

$\mathcal{F}(s, \vec{\sigma}) = (\Phi_1(\gamma_\sigma(s)), \dots \Phi_n(\gamma_\sigma(s)))$

Imamo enačbo $\mathcal{F}(s, \vec{\sigma}) = \vec{l}$

Z izrekom o implicitni preslikavi dobimo $\vec{\sigma} = \vec{\sigma}(s)$ v okolici $(0,0)$. $\mathcal{F}(0,0) = l$ $\sigma(0) = 0$

$$\frac{d\mathcal{F}}{d\sigma}(0,0) = \begin{bmatrix} \frac{dF_1}{d\sigma_1} & \dots & \frac{dF_1}{d\sigma_n} \\ \vdots & & \vdots \\ \frac{dF_n}{d\sigma_1} & \dots & \frac{dF_n}{d\sigma_n} \end{bmatrix}(0,0)$$

$$\frac{d\mathcal{F}_i}{d\sigma_j}(0, \vec{\sigma})\Big|_{\vec{\sigma}=0} = \frac{d}{d\sigma_j}\Big|_{\vec{\sigma}=0} \Phi_i(u + \sigma_1 w_1 + \dots + \sigma_n w_n) =$$

$$= (D_u\Phi_i) \cdot w_j = \delta_{ij} \implies D^{\sigma}_{(0,0)}\mathcal{F} = I \Rightarrow \vec{\sigma} = \vec{\sigma}(s)$$

$\Rightarrow \gamma : s \mapsto u + sv + \sigma_1(s)w_1 + \dots + \sigma_n(s)w_n$ je dobro definiran

$\gamma'(0) = v + \sigma_1'(0)w_1 + \dots + \sigma_n'(0)w_n$

$$\frac{d}{ds}\Big|_{s=0} \mathcal{F}(s, \vec{\sigma}(s)) = \begin{bmatrix} \frac{d}{ds}\big|_{s=0} \Phi_1(\gamma(s)) \\ \vdots \\ \frac{d}{ds}\big|_{s=0} \Phi_n(\gamma(s)) \end{bmatrix} =$$

$$= \begin{bmatrix} D_u\Phi_1 \cdot \gamma'(0) \\ \vdots \\ D_u\Phi_n \cdot \gamma'(0) \end{bmatrix} = 0 \rightsquigarrow \gamma'(0) \in \bigcap_{i=1}^{n} \ker \Phi_i$$

■

Lema za splošnejši problem

- $\varphi_1 \dots \varphi_n$ lin. neodvisni lin. funkcionali $U \to \mathbb{R}$
- $\bigcap_{i=1}^{n} \ker \varphi_i \subseteq \ker \varphi$ za nek $\varphi : U \to \mathbb{R}$

Potem obstajajo λ_i, da velja $\varphi = \sum_{i=1}^{n} \lambda_i \varphi_i$

Dokaz: Naj bodo $\{v_1 \dots v_n\} \subseteq U$, da velja

Definirajmo $w = u - \sum_{i=1}^{n} \varphi_i(u) \cdot v_i$

$$\varphi_k(w) = \varphi_k(u) - \sum_{i=1}^{n} \varphi_i(u) \cdot \varphi_k(v_i) = \varphi_k(u) - \varphi_k(u) = 0$$

$$\Rightarrow w \in \bigcap_{i=1}^{n} \ker \varphi_k$$

$$\Rightarrow w \in \ker \varphi$$

$$0 = \varphi(w) = \varphi(u) - \sum_{i=1}^{n} \varphi_i(u) \cdot \overbrace{\varphi(v_i)}^{\lambda_i}$$

$$\Rightarrow \varphi(u) = \sum_{i=1}^{n} \lambda_i \varphi_i(u)$$

$\blacksquare$

Splošnejši problem

- $\mathcal{L}: U \to \mathbb{R}$ funkcional
- $\Phi: U \to \mathbb{R}$ vez
- $u \in U$; $\Phi_i(u) = l_i$ $D_u \Phi_i \neq 0$

Potem je u lokalni ekstrem $\mathcal{L}$ pri pogojih $\Phi_1 \dots \Phi_n$

$\Leftrightarrow \exists \lambda_i \in \mathbb{R}.\ D_u(\mathcal{L} - \sum \lambda_i \Phi)$

Dokaz: $V = \{u \in U ; \forall i.\ \Phi_i(u) = l_i\}$

za $v \neq u$

Recimo da je u lokalni ekstrem $(\forall v \in V.\ \mathcal{L}(u) < \mathcal{L}(v))$

Naj bo $\gamma : [\varepsilon, \varepsilon] \longrightarrow V$ pot $\gamma(0) = u$

$$\frac{d}{ds}\Big|_{s=0} \mathcal{L}(\gamma(s)) = (D_{\gamma(0)} \mathcal{L})\, \gamma'(0) \quad \text{za } \forall \gamma$$

$$v \in \ker D_u \mathcal{L} \Leftrightarrow (D_u \mathcal{L}) v \Leftrightarrow \begin{bmatrix} D_u \Phi_i \\ \vdots \\ D_u \Phi_i \end{bmatrix} v = 0$$

$$\Leftrightarrow v \in \bigcap_{i=1}^{n} \ker D_u \Phi_i$$

$$\Leftrightarrow \ker D_u \mathcal{L} = \bigcap_{i=1}^{n} \ker D_u \Phi_i$$

$$\Leftrightarrow D_u \mathcal{L} = \sum_{i=1}^{n} D_u \Phi_i \cdot \lambda_i$$

Imamo torej u je ekstrem $\Leftrightarrow D_u(\mathcal{L} - \sum_{i=1}^{n} \lambda_i \Phi_i) = 0$ za neke λ_i. Torej ima u ekstrem $\Leftrightarrow$ ko ima ekstrem $\mathcal{L} - \sum_{i=1}^{n} \lambda_i \Phi_i$